

NEURAL BELLMAN OPERATORS

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This paper develops Neural Bellman Operators for constructive policy evaluation and feasible improvement. An own-future continuation retains action witnesses and an explicit policy-loss account. In a nonlinear investment economy, recursive Bellman queries replace the state lattice while a trained selector proposes where to obtain information. We derive a computable pre-query certificate, a loss that bounds additional verification work, and a fixed-feature ReLU readout with a certified optimization gap. The same original-optimum account extends to two controls and two continuous innovations. A source-frozen study compares ten selectors, eight fitting seeds, prospective cached reuse, and actual callable policies under the continuous law. Complete work, certificate progress and policy cost are reported separately. Reuse savings depend on the task and comparator; expected policy-cost differences remain statistically unresolved. The purchase criterion combines these measured objects without assuming neural superiority or equating a tighter certificate with an economic gain.

KEYWORDS: Dynamic programming, Neural approximation, Policy certification, Information acquisition, Computational economics.

1. INTRODUCTION

A continuation approximation matters because it helps produce an economic decision. Prediction error, the loss of the implemented policy and the work needed to certify that loss are different objects. This paper connects them within a constructive Neural Bellman Operator (NBO). The economic target is a feasible complete policy under the original law, not an action that is optimal only for a fitted surrogate.

The first construction evaluates each continuation against its own fitted future and retains feasible action witnesses. Its affine-ReLU realization and primitive error account do not require a successful nonconvex optimizer. The second imple-

mentation queries the original Bellman problem recursively, only at states needed by its current action search. A trained selector proposes where to query; outward lower and upper bounds determine whether information is sufficient. These are two implementations within the same NBO development, not two encodings counted as independent methods.

The nonlinear investment economy has state-dependent capacity and continuous innovations. Its primitive curvature controls the convexity defect of the bilinear transition. This yields Lipschitz and semiconcavity bounds for the true Bellman value, and hence continuous-action certificates without a tensor state table. The trained continuation or selector is not assumed convex. Boundary queries, protected refinement, acquired-state feasibility and an explicitly costed rational recovery determine the complete policy’s original-optimum guarantee.

The new operational question is whether a proposed action can be licensed before purchasing its Bellman value. Strong convexity gives an upper chord from two existing endpoint bounds. Subtracting the global action-search lower bound produces a computable regret certificate. If that certificate closes, the actual rounded proposal can be returned without another query. If it does not close, protected conventional refinement remains available. This criterion replaces an unobserved distance to the minimizing action by information present in the numerical transcript.

A corresponding margin loss measures unresolved verification work. Its achieved value bounds additional endpoint queries on a specified context catalogue. A separate validation result states the distributional conclusion and the required independence assumptions. We also give a fixed-feature ReLU readout whose certificate-aware objective is convex and whose optimization error is bounded by a computable box dual gap. This constructive training result does not assert convergence of unrestricted hidden-layer fitting. The numerical comparison retains its separately frozen L-BFGS, quadratic and other selectors, and evaluates their actual certificate progress rather than equating action-label fit with verification savings.

The empirical design tests the complete procedure. Ten selectors receive identical kernels and screens, including linear, spline, nearest-action, derivative-based and simple analytic controls. Eight predetermined fitting seeds separate training vari-

ability from two clock repetitions. A prospective reuse experiment loads a saved model for twenty-four new workloads and deliberately declines a stale acquisition identifier. Fresh common-path inference evaluates the actual callable controllers at every date under the original continuous law. A two-control, two-innovation extension reaches four dates and reports the resulting descendant cost and memory instead of inferring horizon scalability from state dimension alone.

An independent rational check also detected a path-accounting error in the preceding execution: mutable lower and upper cumulative-cost endpoints shared an array. We retained that execution, repaired the accumulator and froze a complete new run of the same catalogue. The random indices are unchanged to isolate the correction, so repeated streams are not counted as new independent observations. The active tables use corrected path enclosures and their own complete process clocks. This separates a numerical repair from a change in model, target or selected seed.

The resulting evidence distinguishes three conclusions. The query procedure returns certified policies without a stored state lattice. Learning can change the amount and location of verification work, and a saved selector can be assessed under actual reuse rather than an assumed amortization ratio. Neither conclusion alone identifies lower expected economic cost or representation-specific superiority. The purchase decision therefore uses observed work, direct policy-cost intervals and the declared price of computation as separate inputs.

Relation to existing methods. Lipschitz extension and distance transforms underlie the constructive envelope (McShane, 1934, Felzenszwalb and Huttenlocher, 2012). Approximate and safe policy iteration supply established error and performance analyses (Munos, 2003, Pirodda et al., 2013, Scherrer et al., 2015), while Yang (2020) studies convex dynamic-programming approximation. Prediction-dependent warm starts and learned optimization-based control also have an established literature (Sakaue and Oki, 2023, Sambharya, Hall, Amos, and Stellato, 2023, Agrawal, Barratt, Boyd, and Stellato, 2020). We do not claim those general principles. The additional chain here runs from the original nonlinear economic primitives to an acquired-policy certificate, a pre-query action decision, an operational training loss and

complete measured work. The verifier is method-neutral; the constructive neural representation and the empirical value of a fitted selector are distinct claims.

Reading path. Sections 2–4 retain the original operator, constructive witness theorem and economic regularity. Sections 5–7 establish the recursive query and safeguarded selection. Sections 8 and 9 give the operational training and multivariate query results. Section 10 states the inference contract and Section 11 gives the corrected complete study. The technical supplement contains the inherited proofs and detailed records. The preceding full article, supplement and broader development remain unchanged in the repository; their diffusion, recursive-preference and game results retain their own hypotheses.

2. THE ECONOMIC OPERATOR AND ITS POLICY ACCOUNT

Consider dates $t = 0, \dots, T$ and an invariant compact state domain $X \subset \mathbb{R}^d$. At date t , the action belongs to a nonempty compact set $A_t(x)$. The transition is $F_t(x, a, Z_{t+1})$, the current cost is $c_t(x, a)$, and the terminal cost is $g(x)$. Write $B_0 = 1$ and $B_t = \prod_{j < t} \beta_j$, with $\beta_j > 0$. An implemented policy includes its observation rule, action selection and feasible numerical realization, not just its stored parameters.

For a continuation v , define

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z))), \quad \mathcal{T}_t v(x) = \inf_{a \in A_t(x)} Q_t(v; x, a). \quad (1)$$

Throughout the deterministic comparison, the evaluations ρ_t are monotone and cash invariant on a common domain containing all functions and constant translates used below. They are therefore nonexpansive in the supremum norm. Continuous objectives, effective compact feasible correspondences and measurable minimizers, or measurable arbitrarily near-minimizers with their error charged, are assumed. The stipulated conditional recursion gives Bellman comparison against all admissible adapted controls. In the experiments ρ_t is conditional expectation, costs are bounded and innovations have their original continuous uniform law. The direct-cost inference in Section 10 is specific to expectation; it is not automatically an inference theorem for nonlinear recursive utility or games.

A Neural Bellman Operator is a construction and evaluation procedure for (1). It forms targets using its *own* fitted f_{t+1} , returns a neural continuation and a feasible policy, and records an economic error allowance. It does not receive the optimal continuation as a training label. Arbitrary trained networks can be evaluated by the first result below. The subsequent constructive backend specifies how a network and its actor are obtained without assuming successful nonconvex optimization.

2.1. Action-dependent errors and complete-policy loss

Let $H(a)$ be a true current cost and $\hat{H}(a)$ its continuation approximation. If $\hat{a}$ is δ -optimal for $\hat{H}$ on the same feasible action set, then

$$H(\hat{a}) - \inf_a H(a) \leq \delta + \text{osc}_a \{\hat{H}(a) - H(a)\}. \quad (2)$$

Indeed, compare $\hat{a}$ with an arbitrarily near-optimal true action and cancel the approximate objectives. The infimum limit gives the result. Additive continuation levels cancel; action-dependent discrepancies do not. Under a change in future policies, the continuation must be transported to the new economic problem before this inequality licenses reuse. The full development edition retains the original quantitative transport, recursive-domain and economic-application results. No transport bound is presumed valid after an unaccounted change of constraints or preferences.

PROPOSITION 1—One-sided policy sandwich: *Suppose, uniformly in state,*

$$f_t - \mathcal{T}_t f_{t+1} \leq u_t, \quad \mathcal{T}_t^\pi f_{t+1} - f_t \leq v_t, \quad \ell_T \leq f_T - g \leq u_T.$$

Then the actual feasible policy satisfies

$$0 \leq J_0^\pi - V_0^* \leq \sum_{t < T} B_t(u_t + v_t) + B_T(u_T - \ell_T). \quad (3)$$

The bound is invariant to date-specific constant shifts of the fitted continuations, provided the associated residual accounts are shifted consistently.

The proof is in Supplement, Section 1.1.

A sharper valid residual account for the same continuation and the same deployed actor sharpens knowledge of the policy. It does not retrain that policy or change its realized cost. Conversely, subtracting two upper bounds in (3) does not identify the difference in actual costs. These distinctions govern both the historical comparisons and the new experiments.

3. CONSTRUCTING AND COMPILING A FEASIBLE NEURAL POLICY

Assume the primitive objective has, for feasible state-action pairs, state modulus $A_t^Q = C_{x,t} + \beta_t F_{x,t} L_{t+1}$ and action modulus $D_t = C_{a,t} + \beta_t F_{a,t} L_{t+1}$. Let

$$d_H(A_t(x), A_t(z)) \leq K_t^A \|x - z\|_1, \quad L_t = A_t^Q + D_t K_t^A, \quad L_T = L_g.$$

The Hausdorff condition and the primitive moduli imply that $\mathcal{T}_t f_{t+1}$ is L_t -Lipschitz whenever f_{t+1} is L_{t+1} -Lipschitz. All moduli and numerical integration procedures are supplied by the economic primitives, not fitted from successful outcomes.

At state nodes $x_{t,i}$ with covering radius h_t , form effective feasible k_t -nets. Evaluate their own-future objectives with certified error at most e_t . Minimize the computed labels and retain the corresponding *original* action $a_{t,i}$. A Borel repair $\Gamma_{t,i}(x)$ is feasible at x and satisfies

$$\|\Gamma_{t,i}(x) - a_{t,i}\|_1 \leq K_t^A \|x - x_{t,i}\|_1 + \zeta_t. \quad (4)$$

The allowance ζ_t concerns displacement; feasibility itself may not fail. Define

$$f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}, \quad i_t(x) = \min \arg \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}. \quad (5)$$

Return $\pi_t(x) = \Gamma_{t,i_t(x)}(x)$. Terminal labels approximate g with error e_T and use the same cone construction with modulus L_g .

THEOREM 2—Own-future construction with feasible witnesses: *The functions in (5) have exact affine-ReLU realizations and Lipschitz moduli L_t independent of label error. The returned policy is feasible and measurable. It satisfies*

$$-e_t \leq f_t - \mathcal{T}_t f_{t+1} \leq D_t k_t + e_t + 2L_t h_t, \quad (6)$$

$$\mathcal{T}_t^\pi f_{t+1} - f_t \leq e_t + D_t \zeta_t,$$

and hence

$$J_0^\pi - V_0^* \leq \sum_{t < T} B_t(D_t k_t + 2e_t + 2L_t h_t + D_t \zeta_t) + B_T(2e_T + 2L_g h_T). \quad (7)$$

With effective covers, repairs and certified queries at the prescribed accuracy, allocating each nonnegative term a share of a positive economic tolerance gives a terminating finite construction. A fixed numerical precision or a capped finite experimental ladder need not contain that allocation.

The proof is in Supplement, Section 1.2.

The Lipschitz-extension principle is classical (McShane, 1934). A native min-plus envelope and its exact ReLU circuit, using the same labels, witnesses and ties, are one method with two encodings. They return the same function and policy. Neither Theorem 2 nor a faster representation proves a uniquely neural capability or a training-success probability for Adam or L-BFGS. The mathematical increment is the complete feasible, own-future construction and implementation account; the empirical question is its work and policy value relative to conventional generators.

3.1. Original-owner compilation on a tensor cover

On a tensor cover $\mathcal{X} = \prod_{j=1}^d \{x_{j,0}, \dots, x_{j,n_j}\}$, put $S = \prod_j (n_j + 1)$. For arbitrary labels y_i , including inconsistent inexact labels, compute at each grid node v

$$(z_v, o_v) = \min_{\text{lex}, i} (y_i + L\|v - x_i\|_1, i).$$

The index o_v is the original label owner, not the transformed node v .

THEOREM 3—Continuous original-owner identity: For a closed tensor cell C with corners $\mathcal{V}(C)$, and $x \in C$,

$$f(x) = \min_{v \in \mathcal{V}(C)} (z_v + L\|x - v\|_1), \quad (8)$$

$$(f(x), i_*(x)) = \min_{\text{lex}, v \in \mathcal{V}(C)} (y_{o_v} + L\|x - x_{o_v}\|_1, o_v).$$

Coordinatewise forward and backward relaxations compute (z, o) in $O(dS)$ additions and comparisons and $O(S)$ storage. After locating the cell, a query uses $O(d2^d)$ arithmetic

operations. Exact rational comparisons preserve all ties, including $L = 0$. Using the original owner's repaired action preserves the uncompiled policy, its cost and its certificate exactly.

The proof is in Supplement, Section 1.3.

The nodal algorithm is the classical distance transform (Felzenszwalb and Huttenlocher, 2012); preserving the original action through continuous off-grid queries is the policy obligation. If there are $J_{t,i}$ node actions and M_t integration representatives, write $Q_t = M_t \sum_i J_{t,i}$. Literal dense cone evaluation costs $O(\sum_t dS_{t+1}Q_t)$ in representation work. Compilation replaces this by $O(\sum_t [dS_{t+1} + Q_t d2^d])$, plus cell location, primitive evaluation, integration, feasible repair, certificate and output. Rational bit lengths and state boxes crossing many cells incur additional costs. There is no dimension-free complexity assertion.

3.2. Acquired states and conventional actors

A b -bit coordinate sensor reports a dyadic cell and its midpoint $\hat{x}$. On the unit cube, $\|x - \hat{x}\|_1 \leq r(b) = d2^{-b-1}$. For the investment capacity $\kappa(x)$ below, clip the original action to the smallest capacity throughout that cell and round down to an action quantum Δ_t . This is feasible at every state in the cell. Relative to an exact true-state repair, its displacement is at most $2K_t^A r(b) + \Delta_t$. Transporting the witness cone from $\hat{x}$ to x adds at most $2L_t r(b)$, plus any certified selector objective error ν_t . Thus the implemented-policy account adds

$$\sum_{t < T} B_t \{ (2L_t + 2D_t K_t^A) r(b) + D_t \Delta_t + \nu_t \}. \quad (9)$$

No continuity of the discrete owner selector is assumed. At uncertain numerical comparisons the simulator retains all possible actors, rather than choosing a favorable one.

For conventional FVI the continuation is multilinear on its realized tensor grid. Its actual modulus is bounded by the largest coordinate edge slope. A nearest-node actor uses a feasible node minimizer and true-state repair. Let h_t be the largest half-cell ℓ^1 radius, L_t^{gr} the Bellman graph modulus, and e_t the node query allowance. Its optimal residual upper bound is $D_t k_t + e_t + L_t^{\text{gr}} h_t$. Its selected-policy upper bound

is $e_t + E_t$, where

$$E_t = \sup_x \{y_{i(x)} + L_t^{\text{gr}} \|x - x_{i(x)}\|_1 - f_t(x)\}.$$

Partitioning cells by nearest-node boundaries makes the expression separately affine in each coordinate. Its maximum is therefore at a partition vertex; the implementation encloses all such vertices. Both uniform and curvature-refined FVI receive this sharper one-sided account, not an artificially doubled baseline. Their acquired-state allowance is derived separately in (37).

4. ORIGINAL-OPTIMUM ACCURACY IN THE INVESTMENT ECONOMY

The exact witness theorem answers an implementation question conditional on a continuation. An economist also needs to know how close the complete implemented policy is to the original optimum. We now establish that connection in the investment economy itself. The argument exploits a property not used by the earlier cell-range calculation: despite the bilinear transition, the Bellman objective remains jointly convex. This permits a second-order interpolation account and transfers verified nodal decisions to an implementable policy over the entire state domain. It does not assume that a fitted neural continuation is convex.

Relation to the preceding two-control query. The earlier query experiment used the fixed shared capacity $1/8$. The complete dynamic extension below uses the original state-dependent capacity $c(x)$, with the retained exposure columns, action-cost interaction and second shock. Its nesting claim concerns the original scalar economic model; it is not an assertion that a full dynamic policy was already present in the earlier isolated query.

4.1. Primitives and Bellman regularity

Let $d \geq 2$ be even, let indices be cyclic, and set $\beta = 15/16$. Write

$$S(x) = \frac{2}{d} \sum_j (x_j - 5/8)^2 + \frac{1}{4d} \sum_j (x_j - x_{j+1})^2 + 2 \left(\frac{1}{2} - \frac{2}{d} \sum_j x_j \right)_+^2,$$

$$g(x) = S(x) + \frac{2}{d} \sum_j (x_j - 5/8)^2, \quad c(x) = \frac{1}{8} + \frac{1}{8d} \sum_j x_j.$$

The original scalar control has $0 \leq a \leq c(x)$, running cost $S(x) + a^2 + 4a^4$, and transition

$$F_j(x, a, z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1 - x_{j+1})}{16} + B_j a + s_j z,$$

where $(B_j, s_j) = (1/2, 1)$ at alternating coordinates and $(1/4, -1)$ at the others, and z is uniform on $[-1/32, 1/32]$.

For the retained two-control extension, $a_1, a_2 \geq 0$, $a_1 + a_2 \leq c(x)$, the second exposure column interchanges $1/2$ and $1/4$, and the action cost is

$$R(a) = a_1^2 + a_2^2 + 4(a_1^4 + a_2^4) + a_1 a_2 / 4.$$

An independent common-direction uniform innovation of radius $1/64$ is added. This is the previously specified extension, nested at $a_2 = 0$ when that innovation is disabled. Both models preserve $[0, 1]^d$. Let $m \in \{1, 2\}$ denote the number of controls and define

$$Q_t^*(x, a) = S(x) + R(a) + \beta \mathbb{E} V_{t+1}^*(F(x, a, Z)),$$

$$V_t^*(x) = \min_{a \in A(x)} Q_t^*(x, a), \quad V_T^* = g.$$

All costs and laws in this display are the economic primitives, not a fitted surrogate.

THEOREM 4—Regularity of the original Bellman problem: *For either control specification and every finite horizon, V_t^* is convex and continuously differentiable in the interior, with the corresponding one-sided extension on the cube. It is G_t -Lipschitz in the ℓ^1 norm and M_t -semiconcave in the Euclidean norm, where valid constants are*

$$G_T = 10/d, \quad G_t = \frac{1037}{128d} + \beta \frac{47}{64} G_{t+1},$$

$$M_T = 26/d, \quad M_t = \frac{22}{d} + \frac{21}{256d} + \beta \left(\frac{9}{16} M_{t+1} + \frac{1}{8} G_{t+1} \right).$$

In particular, $G_t \leq 27/d$ and $M_t \leq 54/d$. The function Q_t^* is jointly strongly convex on its convex feasible graph, with state curvature at least $107/(128d)$ and action curvature at least $7/4$.

A valid semiconcavity constant along each coordinate line is

$$D_T = \frac{9}{d} + \frac{16}{d^2},$$

$$D_t = \frac{5}{d} + \frac{16}{d^2} + \frac{21}{256d^2} + \beta M_{t+1} \left(\frac{85}{256} + \frac{23}{256d} \right).$$

Consequently, for the positive multilinear interpolant I_n on a mesh of spacing $1/n$,

$$0 \leq I_n V_t^*(x) - V_t^*(x) \leq \kappa_t, \quad \kappa_t = \frac{dD_t}{8n^2}. \quad (10)$$

The proof is in Supplement, Section 1.4.

The theorem is a structural statement about these economic primitives. It is not a claim that every controlled diffusion, recursive preference specification, or game in the complete development edition has a convex Bellman problem. Nor does it require a signed-weight neural fit to preserve convexity. Convex approximation of dynamic programs has an established literature; the additional work here is to verify the needed curvature for the original nonlinear capacity-constrained model and connect it to the actual NBO actor and complete error account.

4.2. Certified nodal Bellman bounds

Suppose nodal arrays satisfy $l_{t+1} \leq V_{t+1}^* \leq u_{t+1}$ at all grid vertices. Equation (10) gives the functions $I_n l_{t+1} - \kappa_{t+1}$ and $I_n u_{t+1}$ as valid lower and upper continuations. The lower function need not itself be convex; convexity is used for the unknown true value and for the comparison argument, not asserted of arbitrary endpoint arrays.

Divide each uniform innovation component into q equal bins and use its conditional mean. Convexity gives a lower quadrature bound for the true continuation. Semiconcavity gives an upper error

$$\zeta_{t+1} = \frac{M_{t+1}}{6q^2} \sum_{r=1}^s \rho_r^2 \|s_r\|_2^2, \quad (11)$$

where $s = 1$ or 2 , $\rho_1 = 1/32$, and $\rho_2 = 1/64$ in the extension. The expectation remains the original continuous law. At the terminal date g is evaluated directly, so no terminal interpolation allowance is needed.

At a vertex v , let $\mathcal{A}_A(v)$ be the capacity simplex lattice with spacing $c(v)/A$, including all its faces. For $m = 1, 2$, a minimizer can be represented by neighboring lattice vertices whose coordinate variances sum to at most $mc(v)^2/(4A^2)$. A valid action smoothness constant is

$$H_t = \frac{21}{4} + \beta M_{t+1} \|B\|_2^2, \quad \|B\|_2^2 = \frac{5d}{32} \ (m=1), \quad \|B\|_2^2 = \frac{9d}{32} \ (m=2).$$

Taylor's inequality and that barycentric representation imply

$$0 \leq \min_{a \in \mathcal{A}_A(v)} Q_t^*(v, a) - V_t^*(v) \leq \delta_t := \frac{mH_t}{128A^2}. \quad (12)$$

The linear Taylor terms cancel even when the optimum lies on a face. This is an error against the continuous feasible optimum, not only against the finite action menu.

Let $q_t^-(v, a)$ and $q_t^+(v, a)$ include all interpolation, quadrature and arithmetic errors and enclose $Q_t^*(v, a)$. Then

$$l_t(v) = \max\{0, \min_{a \in \mathcal{A}_A(v)} q_t^-(v, a) - \delta_t\}, \quad u_t(v) = \min_{a \in \mathcal{A}_A(v)} q_t^+(v, a) \quad (13)$$

are valid nodal Bellman bounds. The zero truncation uses the nonnegative economic costs. Extra neural or conventional proposals may improve the upper side but do not remove the lower action-cover allowance. Positive interpolation, averaging and minimum maps are monotone and nonexpansive; the separate allowances in (10)–(12) are therefore propagated, rather than being discarded because an endpoint calculation contains the truth.

4.3. From a fitted witness to the complete implemented policy

Let $a_t(v) \in A(v)$ be the actual action returned at each vertex by any generator. It may be selected by the exact neural witness, a conventional fitted continuation, or their common verified fallback. Write $w_v(x)$ for the multilinear weights and

set $\bar{\pi}_t(x) = \sum_v w_v(x) a_t(v)$. Affineness of capacity makes this action feasible. Let $q_t^{\text{sel},+}(v)$ be a verified upper bound on its original-optimum continuation value $Q_t^*(v, a_t(v))$, using the same lower and upper Bellman reference for all generators.

THEOREM 5—Actual-policy, whole-domain certificate: *Set*

$$e_t = \max_v \{q_t^{\text{sel},+}(v) - l_t(v)\} + \kappa_t.$$

Suppose the implemented policy $\hat{\pi}_t$ is feasible and differs from $\bar{\pi}_t$ by at most b_t in action ℓ^1 norm, uniformly over the state cube. Define

$$\Lambda_t = \frac{13}{16} + \beta \frac{3d}{8} G_{t+1}, \quad D_T^\pi = 0, \quad D_t^\pi = e_t + \Lambda_t b_t + \beta D_{t+1}^\pi.$$

Then, at every true state and date,

$$0 \leq J_t^{\hat{\pi}}(x) - V_t^*(x) \leq D_t^\pi.$$

This statement is conditional on the actual recorded policy, not on success of its fitting algorithm. It applies equally to a pure fitted policy and to a policy augmented by verified common candidates.

The proof is in Supplement, Section 1.5.

In the execution, observation is rounded down at forty bits. An acquired state $y \leq x$ has no greater capacity than the true state. Each interpolated action is then rounded down at twenty bits using the lower endpoint of its arithmetic enclosure. If L_{ij} bounds the derivative of action coordinate i with respect to state coordinate j , and η bounds native interpolation error, a valid implementation allowance is

$$b_t \leq 2^{-40} \sum_{i,j} L_{ij} + m(2^{-20} + 2\eta + \eta_{\text{out}}).$$

Here L_{ij} is computed from the maximum adjacent nodal difference times n , and η_{out} covers endpoint subtraction. All these numbers are recorded outward. This new barycentric acquired actor is explicitly distinguished from the earlier discontinuous cell-constant actor; an old actor is not silently replaced during a purported fixed-policy revalidation.

4.4. Arithmetic, work, and scope

The native interpolation kernel forms nonnegative tensor weights and a weighted sum of at most 2^d stored binary64 endpoints. It is compiled without reassociation or floating-point contraction. For $K = (3d + 4)2^d + 32$, an explicit conservative absolute allowance is

$$\eta = \frac{K2^{-53}}{1 - K2^{-53}} \max_v |z_v| + K2^{-1022}.$$

The final term conservatively covers underflow. Input coordinates, affine indexing, finite endpoints and the binary64 format are checked. Roundoff expansion and the ordinary interval cost arithmetic remain separate from the analytic approximation bounds. Exact rational fixtures compare the native sum with its mathematical tensor interpolant.

With $N = (n + 1)^d$ vertices, $K_A = A + 1$ scalar actions or $K_A = (A + 1)(A + 2)/2$ two-control actions, the reference calculation uses $O(TNK_A q^s 2^d d)$ elementary work under the explicit tensor implementation. Stored reference values use $O(TN)$ words and a policy uses $O(TNm)$ words. Streaming batches avoid a state-pair table. Proposal fitting, exact rational witness work, actor transfer, serialization, independent cost inference and compilation are additional costs, not absorbed into this bound. The analytic errors are second order in state, action and innovation resolution; the state-count and tensor-evaluation factors still depend exponentially on dimension. The result supplies a matched original-optimum comparison without asserting that these costs have disappeared.

5. NEURAL BELLMAN QUERIES WITHOUT A STATE LATTICE

The preceding construction makes numerical error accountable, but a complete tensor reference can absorb the benefit of a fitted continuation. We therefore change the computational organization, not the economic problem. A Neural Bellman selector proposes an action at the state actually acquired. A recursive interval query then bounds the original Bellman value at that state and checks the proposed action against the entire continuous feasible interval. No conventional policy is precomputed at all states. The controller is the resulting callable algorithm, not the unverified proposal alone.

The distinction matters for attribution. Learning may reduce the work needed to locate a useful action. It does not supply the inequalities used to certify that action. The adaptive conventional comparator receives the same inequalities and the same original-optimum target. The trained value-fit policies in the preceding study remain distinct from the action-query predictor introduced here.

5.1. A curvature lower bound on an adaptive action partition

Use the original scalar economy of Section 4, with r dates remaining. Write V_r for its optimal value and $Q_r(x, a)$ for the value of a first action followed by the optimal future. Let G_r, M_r be the primitive constants in Theorem 4, indexed by remaining horizon, and put

$$H_r = \frac{21}{4} + \beta M_{r-1} \frac{5d}{32}, \quad \Lambda_r = \frac{13}{16} + \beta G_{r-1} \frac{3d}{8}.$$

Thus H_r is an upper curvature bound in the scalar action, and Λ_r bounds the absolute action derivative. Neither depends on a fitted model.

LEMMA 6—Parabolic minorants: *Suppose f is H -semiconcave on $[l, u]$, and $L_l \leq f(l)$, $L_u \leq f(u)$. For $s \in [0, 1]$,*

$$f(l + s(u - l)) \geq (1 - s)L_l + sL_u - \frac{H}{2}(u - l)^2 s(1 - s) =: p(s).$$

Consequently $\min_{[0,1]} p$ is a lower bound for the minimum of f on the interval. With $K = H(u - l)^2/2$ and $D = L_u - L_l$, this lower bound is L_l if $D \geq K$, L_u if $D \leq -K$, and otherwise

$$\frac{L_l + L_u}{2} - \frac{K}{4} - \frac{D^2}{4K}.$$

The endpoints L_l, L_u need not themselves belong to a convex interpolant. Increasing K preserves the lower-bound property.

The proof is in Supplement, Section 1.6.

At an acquired state, retain an ordered partition of the feasible action interval and an outward value bracket at every partition point. The minimum of the parabolic minorants bounds the continuous-action infimum from below. The smallest upper

endpoint supplies an actual feasible witness. A neural or polynomial proposal inserts another point into this partition; it cannot remove the endpoints or any part of the feasible interval. The running lower bound is the maximum of all previously valid global lower bounds, so new information cannot weaken the certificate. The upper bound is the minimum over every evaluated feasible witness.

5.2. Recursive queries under the original continuous law

For an endpoint action with $r > 1$, divide the scalar innovation into q_r equiprobable bins and query V_{r-1} at each conditional-mean next state. If each child returns width at most η_r , the endpoint bracket has width at most

$$\beta\eta_r + \frac{\beta M_{r-1}d}{6144q_r^2} + \omega_r. \quad (14)$$

Here ω_r includes outward primitive arithmetic and the Lipschitz expansion from an interval-valued next state to its queried center. Convexity provides the lower conditional-mean inequality; semiconcavity and the exact conditional variance provide the upper allowance. The innovation remains continuous. There is no state-interpolation term because a child value is queried rather than read from a state lattice.

For a requested local gap δ , the implementation sets $\eta_r \leq \delta/(4\beta)$ and chooses a dyadic q_r so that the second term in (14) is at most $\delta/4$. A separate endpoint-width guard charges arithmetic and the small capacity-rounding remainder. Failure of that guard is a precision failure, not a successful return. The terminal query uses analytic integration instead of recursive quadrature.

THEOREM 7—A state-grid-free NBO service: *Consider the original scalar investment economy. At every recursive query, suppose the outward endpoint and capacity-transfer accounts give width at most $5\delta/8$. Let the lower bound be the running maximum described above. Refine an interval attaining the smallest current minorant whenever the global upper-minus-lower gap exceeds δ . Each new point must lie between one fifth and four fifths of that interval. Arbitrary additional learned proposals may be inserted initially. Then:*

(i) every returned interval $[L_r(x), U_r(x)]$ contains $V_r(x)$, and its returned feasible action $a_r(x)$ satisfies

$$Q_r(x, a_r(x)) - V_r(x) \leq U_r(x) - L_r(x) \leq \delta;$$

(ii) the real-arithmetic refinement terminates. With at most one interior learned proposal, a sufficient point budget is

$$6 + \left\lceil 5c(x) \sqrt{H_r / (3\delta)} \right\rceil; \quad (15)$$

(iii) for the policy that calls this service at every subsequent date, the original-law loss satisfies

$$J_T^\pi(x) - V_T(x) \leq \sum_{t=0}^{T-1} \beta^t \left[\delta_t + \left(8 + \frac{11\beta d G_{T-t-1}}{16} + d G_{T-t} \right) 2^{-b} \right], \quad (16)$$

when the state is acquired downward at b bits per coordinate and every returned action is feasible at that acquired state. All action rounding must precede its endpoint evaluation. The inequalities hold uniformly in the true initial state, not only at the states used in an experiment.

The proof is in Supplement, Section 1.7.

The finite-precision contract is explicit. The executed service uses the width guard, a 512-point cap, and a 32-bit downward action lattice. A precision or cap failure withholds the floating certificate and invokes a separately counted exact-rational enumeration. This fallback constructs values only at the required descendants; it does not load a precomputed conventional policy. The rational fallback chooses dyadic action and quadrature counts until each cover or quadrature allowance is at most one quarter of the request, and requests child width at most one quarter divided by the discount. Provided $\delta > 8\Lambda_r 2^{-32}$ at every recursive level, its exact brackets have width at most three quarters of the request. This proves a finite return for every acquired state in the declared tasks, including states absent from the workload. Rational bit-operation cost is additional work and is not replaced by a unit-cost floating clock. The mathematical point budget is checked against the declared tolerance range; termination in an arbitrary floating implementation

is not inferred merely from the real-arithmetic argument. Fitting convergence is unnecessary for validity: any finite prediction, and even its clipped replacement when numerically invalid, is only a proposed query.

5.3. A certified analytic terminal kernel

The last decision admits an especially strong common comparator. Let $y = F(x, 0, 0)$, $b_0 = 1/2 - 2\text{mean}(y)$, and $\gamma_j = B_j$. Define

$$D(x) = \frac{8}{d} \sum_j \gamma_j (y_j - 5/8) + \frac{1}{2d} \sum_j (\gamma_j - \gamma_{j+1}) (y_j - y_{j+1}).$$

For every even d , the exact terminal action derivative is

$$16a^3 + \left(2 + \frac{41\beta}{32}\right)a + \beta D(x) - 3\beta(b_0 - 3a/4)_+. \quad (17)$$

Each branch is a strictly increasing cubic. Its floating closed-form root supplies only a candidate. The independently evaluated outward derivative and exact-law objective certify that candidate; a safeguarded derivative bracket remains available if the candidate is inaccurate.

For completeness, if f is μ -strongly convex and an outward interval encloses $f'(a)$, the largest compatible projected derivative residual R gives

$$f(a) - \min_{[0,c]} f \leq R^2 / (2\mu).$$

At zero, a nonnegative derivative has residual zero; at capacity, a nonpositive derivative has residual zero. Otherwise the absolute derivative bound is sufficient. This follows by minimizing the strong-convexity supporting quadratic over the feasible interval and, conservatively, over the line. Here $\mu = 2$. Equation (17), its $41/32$ and $9/4$ coefficients, analytic shock moments, and outward derivatives are independently checked against rational formulas. Every method uses this kernel; it is not credited to the neural predictor.

5.4. *Work, prediction, and the economic object*

Let $A_r(\delta)$ be a bound on action endpoints at a query. Excluding training, a scalar operation account obeys

$$W_1(\delta) = O(dA_1), \quad W_r(\delta) \leq CdA_r(\delta) + A_r(\delta)q_r W_{r-1}(\delta/(4\beta)).$$

The constants dG_r, dM_r are bounded independently of d , so the declared endpoint and quadrature budgets need not grow exponentially with the state dimension. Primitive evaluations still cost $O(d)$. This removes the state lattice, not the horizon or shock-tree cost. The executed vectorization batches a query’s innovation descendants. Its largest live state batch is reported; it does not claim the smaller memory of a fully streamed implementation. Increasing the horizon or the shock dimension may still be expensive. The new execution is scalar-control; the earlier full two-control result is retained rather than relabeled as a query-service experiment.

A useful prediction can reduce the endpoint count or return its own action from the initial partition. An inaccurate prediction can instead add work. This is a prediction-dependent algorithm, not a theorem of neural dominance. Learned warm starts have their own literature, including [Sakaue and Oki \(2023\)](#); their discrete-convex setting differs from the original-law Bellman intervals here. Convex control policies and learned optimization parameters are studied by [Agrawal, Barratt, Boyd, and Stellato \(2020\)](#). Our comparison identifies the extra proposal cost and the resulting query transcript rather than claiming priority for learning a warm start.

The purchase is now a complete callable controller with a declared error contract. Its training cost, every subsequent certified query, and its return-time cost belong to the same service. A reduction in queried state storage is economically relevant for repeated decisions only after these costs are counted. The original cost normalization is maintained. Workload seconds and normalized expected-cost loss are reported separately; a price per unit of computation can convert the former to the latter only as an explicitly specified decision-maker parameter.

6. CERTIFICATE-GATED NEURAL QUERIES

A prediction of the optimal action and a prediction of a useful numerical query are different objects. Inserting a learned action before examining an already available

certificate charges an extra continuation query even when the two boundary actions suffice. We instead make the demand for another query a mathematical decision and let a trained NBO selector determine its location only after that demand has been established. The underlying continuation recursion and the economic optimum are unchanged.

At a recursive state, start with the two feasible action endpoints. Compute the global lower bound from the parabolic minorants of Lemma 6, including the capacity remainder, and the smallest upper endpoint. Return immediately when their difference is at most the requested tolerance. Otherwise choose a partition interval with smallest current minorant. A learned action is used only if it lies in the central half of that interval; it then replaces the conventional proposed split. Outside that region the safeguarded parabolic split is used. The rounded split must pass the one-fifth interior check before evaluation. A prediction is evaluated at most once at each recursive state and is reused only as a proposed location. It supplies neither a value endpoint nor a lower bound.

THEOREM 8—Gated prediction and complete-policy validity: *Under the assumptions of Theorem 7, the gated service returns the same original-optimum and actual-policy guarantees for every supplied prediction, independently of fitting error. In real arithmetic, with endpoint-plus-capacity width at most $5\delta/8$, a sufficient number of endpoint evaluations at a state is*

$$A_r(\delta, x) = 4 + \left\lceil 5c(x) \sqrt{H_r / (3\delta)} \right\rceil.$$

Each requested evaluation either reduces the unresolved action partition or terminates the query; no initial interior evaluation is mandatory. If the boundary certificate suffices, no predictor is evaluated at that state. If every prediction is declined at every recursive query, the value endpoints and implemented actions coincide with those of the conventional parabolic service using the same arithmetic and tie rules.

The proof is in Supplement, Section 1.8.

This theorem establishes robustness to the numerical content of a fitted prediction, not that learning is free or always useful. It also does not state that an arbitrary routed transcript has fewer queries than the adaptive conventional transcript. The

executed comparison records both possibilities. Unlike a policy catalogue with a stored reference action at every state, neither service constructs a conventional state lattice before it can return a certified policy.

6.1. State dimension, working memory, and reliability

For the original scalar model, the primitive bounds $dG_r \leq 27$ and $dM_r \leq 54$ imply

$$H_r \leq \frac{3369}{256}, \quad \Lambda_r \leq \frac{1319}{128}, \quad \frac{\beta M_{r-1} d}{6144} \leq \frac{135}{16384}.$$

Thus the action and scalar-innovation budgets at fixed remaining horizon and tolerance are independent of the number of state coordinates. Every primitive evaluation still requires $O(d)$ arithmetic. A width- w one-hidden-layer ReLU prediction on the recorded $d + 5$ features requires $O(dw)$ work and stored parameters; a full quadratic prediction on these features uses $O(d^2)$ terms. Neither cost is included for free in the Bellman recursion.

Let $q_r(\delta)$ be the dyadic quadrature count and let $P_r(d)$ be the cost of one predictor evaluation. Apart from the separately identified exact-arithmetic recovery, a conservative work recursion is

$$W_r(d, \delta) \leq CdA_r(\delta) + P_r(d) + A_r(\delta)q_r(\delta)W_{r-1}(d, \delta/(4\beta)).$$

The predictor term is absent on a boundary-certified query. This expression removes the exponential state lattice, not the horizon or innovation-tree dependence. A depth-first scalar implementation can retain the current action partition and one innovation child at a time. The executed batched implementation instead reports its maximum state batch and process peak memory. A zero count of stored state-lattice nodes does not mean zero storage.

The deterministic certificate is valid for any supplied finite network weights. A failure to fit and return such weights is a separate service event. Our reliability estimand is the return count and worst measured complete work over the preregistered finite seed catalogue, with every repetition retained. Numerical workload paths are not optimizer replications. The real-arithmetic cap and the exact-rational preci-

sion condition establish algorithmic termination at admissible tolerances; recorded processor seconds are descriptive, not a hardware-uniform worst-case bound.

6.2. An explicit economic purchase criterion

The decision maker buys a callable controller rather than a table of successful sample decisions. Fix a physical capital numeraire for the normalized state, a conversion $m > 0$ dollars per unit of the model's loss index, a computation charge $p \geq 0$ dollars per measured second, and a fixed deployment fee $f \geq 0$ dollars. For n uses under a specified initial-state law, the monetary objective is

$$\mathcal{C}(\pi) = nm \mathbb{E}[J_0^\pi(X_0)] + pW(\pi) + f.$$

These are declared preference and service-price parameters, not estimated economic primitives. The policy theorem bounds the expected loss under the original continuous law for every such initial-state law. The finite dyadic workload below measures execution; its average cost is not substituted for that expectation.

If two policies have an independently valid expected-gain interval $[l, u]$ for $\mathbb{E}[J_0^{\pi_0} - J_0^{\pi_1}]$, and their incremental measured service charge is $\Delta = p(W_1 - W_0) + f_1 - f_0$, then the monetary net gain lies in

$$[nml - \Delta, nm u - \Delta].$$

This follows by multiplication by the positive numeraire conversion and subtraction of the same service charge. A positive lower endpoint supports purchase; a negative upper endpoint rules it out at the stated prices. Uniform regret certificates alone do not supply a directional expected-gain interval. They instead support a worst-case accuracy requirement imposed before selecting the least expensive eligible service.

Finally, a forecast of amortized query work is distinguishable from a measured complete service. With a fixed model and exactly repeated query workload, if training requires B operations and deployment requires D_L rather than D_C operations per workload, then learning uses fewer total operations after k repetitions precisely when $D_L < D_C$ and $k > B/(D_C - D_L)$. If $D_L \geq D_C$ and $B \geq 0$, no repetition of that unchanged workload produces an operation saving. This arithmetic

criterion includes training and cannot be applied to unrelated future states without a new assumption. The executed catalogue reports its own full process clocks, not a favorable hypothetical amortization.

7. WHEN A PREDICTED QUERY COMPLETES THE CERTIFICATE

The safeguarded theorem is insensitive to the quality of a supplied prediction. A separate question is when that prediction is sufficient to finish verification. The answer must concern both the selected action and the unresolved lower bound. Proximity to a minimizer alone does not make a coarse lower certificate accurate.

PROPOSITION 9—Witness accuracy and certificate localization: *Let f be a differentiable, H -semiconcave objective with $H \geq 0$ on $[0, c]$, let a^* minimize f , and suppose $|f'| \leq \Lambda$. An evaluated proposal $\hat{a}$ lies in $[0, \underline{c}]$, where $0 < \underline{c} \leq c$ and $c - \underline{c} \leq \Delta$. Each endpoint interval is a valid enclosure of f and has width at most w . Suppose the partition containing $\hat{a}$ covers $[0, \underline{c}]$ and has maximum interval length h . Let L be its parabolic lower certificate with the capacity allowance $\Lambda\Delta$, and let U be the smallest upper endpoint, retaining earlier valid bounds as in the query algorithm. With $e = |\hat{a} - a^*|$ and $\lambda = |f'(a^*)|$,*

$$U - L \leq 2w + \Lambda\Delta + \frac{Hh^2}{8} + \lambda e + \frac{He^2}{2}. \quad (18)$$

For an interior minimizer $\lambda = 0$. At a boundary the linear term cannot generally be removed. Thus the right side at most δ is a sufficient certificate-return condition, not an assumption that a trained prediction meets it.

If old and new valid endpoint records give (L, U) and (L', U') after one additional query, with the running maximum lower bound and minimum upper bound, then

$$(U - L) - (U' - L') = (U - U') + (L' - L) \geq 0. \quad (19)$$

An added prediction may have zero certified progress. The same identities apply to a conventional query.

The proof is in Supplement, Section 1.9.

This proposition isolates what a fitted representation can contribute without crediting it with a conventional solution. It can provide a small-error witness and change

which unresolved interval is refined. The global partition, endpoint arithmetic and capacity account still have to support that witness. The maximum-width sufficient condition is deliberately conservative; the implemented minimum-minorant rule can stop earlier when only a subset of the partition remains competitive.

7.1. A complete accuracy contract and its resource dependence

For T dates and target ϵ , write $S_T = \sum_{t < T} \beta^t$. The recorded service chooses a common local tolerance by rounding $\epsilon / (2S_T)$ to binary64 and verifies the final outward allowance. Under $dG_r \leq 27$, a uniform per-date acquisition allowance at forty bits is

$$\alpha_b = \left(8 + \frac{297\beta}{16} + 27 \right) 2^{-40}.$$

The service certifies $(\delta + \alpha_b)S_T \leq \epsilon$. Each descendant request is $\delta / (4\beta)$, rounded as in the actual code and included in its outward endpoint guard. At the smallest request in the two- and three-date catalogues, the declared thirty-two-bit action quantum satisfies the exact-recovery condition $\delta_r > 8\Lambda_r 2^{-32}$. The supplement checks this inequality with rationals for every horizon and target, rather than inferring it from a successful workload.

For fixed remaining horizon and accuracy, the bounds on action endpoints and scalar-innovation bins are independent of state dimension in this normalized economy. The factors dG_r and dM_r are bounded, while every transition and cost still reads d coordinates. A width- w_θ one-layer ReLU predictor has $O(dw_\theta)$ parameters and evaluation work; a dense quadratic on the same features has $O(d^2)$ terms. These statements identify the absence of a $(n + 1)^d$ stored state table. They do not count an arbitrary-precision rational operation as a unit-cost floating operation and do not suppress the product of descendant query counts as horizon grows.

The two fitted implementations and their conventional controls therefore share one original-optimum contract, but need not share a transcript, a returned policy or a runtime. The completed experiments charge their distinct transcripts. A finite-catalogue reliability statement concerns all specified seeds and repetitions, including fitting warnings. It is not a population statement about optimizer success.

8. OPERATIONAL PREDICTION CERTIFICATES AND READOUT TRAINING

Proposition 9 separates witness accuracy from lower-certificate localization. Its distance to an unknown minimizer is useful for analysis but is not a stopping observable. We now replace that distance by an upper envelope computed from information already purchased. This gives a certificate before an additional Bellman query, a training criterion measured in unresolved verification work, and a constructive neural readout with a computable optimization gap.

8.1. A decision before the next query

Fix a state, a date, and the current action partition. Write $f(a) = Q_t^*(x, a)$, and suppose f is μ -strongly convex on the feasible interval. Let $\bar{f}_l, \bar{f}_u$ be verified upper endpoints at two consecutive actions $l < u$. Let L be the current lower bound for the original feasible infimum, including the capacity and arithmetic allowances. For $a = l + s(u - l)$, define

$$C(a) = (1 - s)\bar{f}_l + s\bar{f}_u - \frac{\mu}{2}(a - l)(u - a), \quad 0 \leq s \leq 1. \quad (20)$$

Every symbol in this expression is either a primitive bound or an existing endpoint. No derivative or action at the unknown optimum occurs.

THEOREM 10—Pre-query closure: *Under the preceding conditions, $f(a) \leq C(a)$ throughout $[l, u]$. Let $a_\Delta \in [l, u]$ be the actual feasible rounded proposal. An outward upper endpoint $\bar{C}(a_\Delta)$ satisfying*

$$\bar{C}(a_\Delta) - L \leq \delta \quad (21)$$

certifies $f(a_\Delta) - \inf_{a \in A(x)} f(a) \leq \delta$ without another Bellman query. If the screen does not close, the protected refinement of Theorem 8 retains its validity and finite query bound. The complete acquired policy therefore retains the original-optimum account of Theorem 7.

Equivalently, the union over partition intervals of the sublevel sets $\{a : C(a) - L \leq \delta\}$ is a computable certified action set. Each constituent is a quadratic sublevel interval intersected with its partition cell. Endpoints obtained numerically must be rounded inward, or the final action must be checked by (21).

PROOF: Strong convexity gives $f((1-s)l + su) \leq (1-s)f(l) + sf(u) - \mu s(1-s)(u-l)^2/2$. Replacing the endpoint values by their upper bounds proves (20). Subtraction of a valid lower bound proves (21). A failed screen adds no endpoint to the partition and alters no lower bound. Subsequent protected refinement is therefore covered by the earlier termination proof. A successful screen supplies exactly the local accuracy premise used in the acquired-policy recursion. Finally, C is a convex quadratic in a , proving the sublevel-set assertion. *Q.E.D.*

The screen is shared by every comparator. In particular, minimizing the quadratic upper envelope is an available conventional proposal. Prediction is valuable only to the extent that its proposed location or evaluation cost improves the observed service. The executed neural rule screens its feasible prediction on the partition interval containing it, even when that location is outside the protected part of the currently critical interval. A failed screen does not authorize an unprotected split.

8.2. A computable loss for unresolved work

A *certificate context* z consists of the state, remaining horizon, tolerance, partition endpoints and their verified bounds. A proposal $a_\theta(z)$ is feasible and is computed without another Bellman query. Define its verified score and a margin loss by

$$s_\theta(z) = \overline{C}_z(a_\theta(z)) - L_z, \quad (22)$$

$$\ell_\tau(z, \theta) = \min \left\{ 1, \frac{(s_\theta(z) - \delta_z + \tau)_+^2}{\tau^2} \right\}, \quad \tau > 0. \quad (23)$$

Let $K(z)$ be a valid cap on further endpoint queries if the screen fails and the protected fallback is followed. This is a count of endpoint queries at the current state. The recursive cost of each endpoint is charged through the earlier descendant-work recursion. On contexts satisfying the declared precision contract, the real-arithmetic and rational-recovery bounds supply finite such caps.

THEOREM 11—Certificate loss and additional work: *If $N_\theta(z)$ is the number of additional endpoint queries after the screen, then*

$$N_\theta(z) \leq K(z) \mathbf{1}\{s_\theta(z) > \delta_z\} \leq K(z) \ell_\tau(z, \theta). \quad (24)$$

Consequently, an achieved margin loss bounds the average additional queries on any specified finite context catalogue, with no unknown minimizing action.

For a distributional assertion, condition on construction of a finite family of M predictors. Suppose $z_1, \dots, z_n$ are independent validation contexts from a declared predictor-independent law, and $0 \leq K(z) \leq K_{\max}$. With probability at least $1 - \alpha$, simultaneously for that family,

$$\mathbb{E}N_\theta(z) \leq \frac{1}{n} \sum_{i=1}^n K(z_i) \ell_\tau(z_i, \theta) + K_{\max} \sqrt{\frac{\log(M/\alpha)}{2n}}. \quad (25)$$

The cost of obtaining and validating contexts and of evaluating the predictor is additional work, not part of N_θ .

PROOF: A successful screen returns without an endpoint query. A failed screen invokes a procedure bounded by $K(z)$, and its margin loss equals one. This proves (24). For each fixed predictor the variable $K(z) \ell_\tau(z, \theta)$ lies in $[0, K_{\max}]$. The bounded-sum inequality of [Hoeffding \(1963\)](#) and a union bound over the conditionally fixed family give (25). Q.E.D.

The context-law requirement matters. A stream of recursively generated contexts can depend on earlier predictions; it is not automatically an independent validation sample for (25). The deterministic inequality (24) remains valid context by context. The recorded action-label mean squared error is not substituted for (23). The new loss measures a certificate's unresolved work rather than resemblance to an approximate action label.

8.3. A constructive ReLU readout with an optimization certificate

The work loss can be trained without invoking a successful hidden-layer optimizer. Fix a nonnegative affine-ReLU feature map $\phi(z) \in \mathbb{R}^p$ with $\sum_j \phi_j(z) = 1$. Linear hat functions on a fixed coordinate partition are one exact construction; averages of such maps give multivariate features. A box-constrained readout $w \in [0, 1]^p$ proposes

$$a_w(z) = l_z + (u_z - l_z) \phi(z)^\top w.$$

The partition interval and features are fixed before optimizing w . Let $\eta_i \geq 0$ cover action rounding and evaluation of the score, so that the computed score of the actual rounded action is at most $C_i(a_w(z_i)) + \eta_i - L_i$. For example, on lattice-aligned endpoints an action Lipschitz bound gives $\eta_i = \Lambda_i \Delta_a$ for downward rounding. Define

$$F(w) = \frac{1}{n} \sum_i K_i(C_i(a_w(z_i)) + \eta_i - L_i - \delta_i + \tau)_+^2 + \frac{\lambda}{2} \|w\|_2^2, \quad \lambda \geq 0. \quad (26)$$

Unlike an action-label objective, this loss directly upper-bounds the additional-work account. It trains the upper witness against the current global lower certificate.

PROPOSITION 12—Certified readout training: *The function F is convex and continuously differentiable on the box. For $g = \nabla F(w)$, put*

$$D(w) = g^\top w - \sum_{j=1}^p \min\{0, g_j\}. \quad (27)$$

Then $0 \leq F(w) - \min_{v \in [0,1]^p} F(v) \leq D(w)$. On the training contexts the average additional endpoint-query count is at most $F(w)/\tau^2$. If $\lambda > 0$, the minimizer w^ is unique and*

$$\|w - w^*\|_2 \leq \sqrt{2D(w)/\lambda}.$$

All quantities in these conclusions can be bounded by outward arithmetic; they are exact rational quantities when the contexts, features and iterate are rational.

PROOF: The chord is a convex quadratic in its normalized action, which is affine in w . Squaring the positive part preserves convexity and gives a continuously differentiable function. The nonnegative weighted sum and quadratic penalty preserve these properties. Convexity gives $F(w) - F(v) \leq g^\top(w - v)$, and the minimum of $g^\top v$ on the box is $\sum_j \min\{0, g_j\}$. This proves (27). Every failed screen has squared positive-part residual at least τ^2 , while every successful screen uses zero further endpoint queries. Summing the corresponding K_i bounds proves the work claim. Strong convexity with modulus λ proves the distance bound. *Q.E.D.*

This is a fixed-feature neural construction, not a convergence assertion for unrestricted nonconvex network fitting. The deposited rational implementation uses finite conditional-gradient steps and reports its achieved loss and dual gap. Its regression examples test the proposition; they are not new investment-policy observations. The service comparison below retains the preregistered L-BFGS and conventional fits, rather than relabeling those models as minimizers of (26). Prediction-dependent optimization has an established literature (Sakaue and Oki, 2023, Sambharya, Hall, Amos, and Stellato, 2023); the additional object here is a computable original-law Bellman certificate that can license an action before another query and supplies a directly auditable training criterion.

9. TWO CONTROLS AND TWO CONTINUOUS INNOVATIONS

The absence of a state lattice does not remove action or innovation dimensions. We extend the query implementation to the previously specified two-control economy of Section 4: $a_1, a_2 \geq 0$, $a_1 + a_2 \leq c(x)$, the interaction cost is $a_1 a_2 / 4$, and the second innovation is an independent common-direction uniform shock. The original scalar model and its experiments are unchanged.

At a queried state, triangulate the feasible action simplex. Let v_0, v_1, v_2 be a triangle's vertices with verified lower endpoints l_0, l_1, l_2 . If $\theta_j(a)$ are its barycentric coordinates, define

$$p(a) = \sum_{j=0}^2 \theta_j(a) l_j - \frac{H}{2} \sum_{j=0}^2 \theta_j(a) \|v_j - a\|_2^2. \quad (28)$$

This is the quadratic $H\|a\|_2^2/2 + b^\top a + c$ interpolating the lower endpoints. Its approximate minimizer is only a proposed support point. For any point $y \in \mathbb{R}^2$, including a numerically displaced point outside the triangle,

$$p(y) + \min_j \nabla p(y)^\top (v_j - y) \leq \min_{a \in \text{co}(v_0, v_1, v_2)} p(a). \quad (29)$$

Thus an outward supporting-plane calculation, not an uncertified quadratic solver, supplies the lower endpoint.

THEOREM 13—Simplicial Bellman query: *For the two-control primitives, (28) is a lower minorant of the true Bellman objective. Taking the smallest valid triangle lower endpoint and the best feasible-action upper endpoint gives a lower–upper certificate for the full continuous simplex. Include the omitted capacity strip and actual action-rounding allowances. Returning when this gap is at most δ gives the same acquired-policy loss recursion as Theorem 7.*

A failed floating triangulation can be replaced by an exact-rational simplex enumeration. Under the stated positive precision slack, this recovery is finite at every finite horizon. With two innovations divided into q bins each, a valid local quadrature error is

$$\zeta = \frac{5\beta M_{t+1}d}{24576q^2}. \quad (30)$$

If A_r bounds the number of action queries at a state with r dates remaining, a work account has the form

$$W_r \leq C_d A_r + A_r q_r^2 W_{r-1}, \quad C_d = O(d), \quad (31)$$

with separate charges for triangulation, arithmetic and output. There is no tensor state-table factor, but the descendant tree remains.

PROOF: Semiconcavity means that $f(a) - H\|a\|_2^2/2$ is concave. Its barycentric inequality, with vertex values replaced by lower endpoints, gives (28). The tangent inequality for the convex quadratic p proves (29); a linear function attains its simplex minimum at a vertex. Taking minima over a cover preserves the lower bound. Each feasible upper witness is an action of the actual model. The local gap premise therefore implies the earlier policy recursion, irrespective of how the triangle or support point was proposed.

For finiteness, use simplex lattice denominator A , with $(A+1)(A+2)/2$ vertices. On triangles of diameter proportional to $c(x)/A$, the curvature error tends to zero as A^{-2} . The endpoint error is reduced by finite descendant accuracy and quadrature refinements. The retained exact-rational recovery contract requires $\delta > 16\Lambda 2^{-32}$ at each recursive call, leaving positive slack after rounding; dyadic enlargement of A and q then terminates. This is a finite arithmetic procedure, not a uniform bound on rational bit cost. For (30), the independent innovation variances

after conditioning on the bins are $\rho_j^2/(3q^2)$, their direction norms squared are d , and the semiconcavity Taylor remainder is $M_{t+1}/2$ times their sum. Substitution of $\rho_1 = 1/32$, $\rho_2 = 1/64$ gives the coefficient. Finally, each action query has q_r^2 descendants, proving (31). Q.E.D.

The executed algorithm uses longest-edge refinement of the lowest triangle and an outward terminal convexity-gap check. Its floating cap is followed by the stated recovery rule, not by an uncertified returned action. The implementation batches descendants but can allocate a descendant array before slicing it into query batches; a reported maximum query batch is therefore not a bound on total live state memory. We report process memory separately. The two-control execution is a verified extension of the NBO query backend, not evidence for a trained two-output selector.

10. FROM RETURNED POLICIES TO ECONOMIC RESOURCE DECISIONS

An all-state upper bound and a comparison of actual policy costs answer different economic questions. We now connect the policies returned by a construction frontier to direct cost inference and then to a choice of implementation resources. The construction data select the policies; independently generated evaluation paths assess them. This separation permits a valid comparison of first-crossing policies with different resolutions.

10.1. *Simultaneous inference after construction*

Let $\mathcal{H}$ denote all construction information: primitive inputs, numerical labels, fitted continuations, returned policies and complete certificate ladders. Conditional on $\mathcal{H}$, fix a finite catalogue of implemented-policy contrasts. Each implementation includes its observation rule, original witness or nearest-node selector, true-state feasible repair and action quantization. For a contrast i , let D_i be the difference in expected discounted costs under a specified initial law. A common-innovation path score evaluates the two policies on their own endogenous trajectories. It is not the difference between two regret bounds.

Suppose the initial and innovation uniforms are partitioned into equal bins. Independent uniform bin indices are drawn, and interval simulation encloses the score for every continuous realization inside the drawn bins. All unresolved state, selector and repair branches remain in the enclosure. If $[A_i, B_i]$ is a deterministic support interval, clip the path endpoints to that support. Write $\underline{m}_i, \bar{m}_i$ for outward sample-mean bounds on the lower and upper endpoint samples and $v_{i,L}, v_{i,U}$ for upper bounds on their unbiased sample variances. For $n \geq 2$ define

$$R_i(v) = \sqrt{2v\ell/n} + \frac{7(B_i - A_i)\ell}{3(n-1)}, \quad \ell \geq \log(4K/\alpha), \quad (32)$$

where K bounds the number of sampled contrasts.

THEOREM 14—Construction-conditional economic intervals: *Under the stated independent-bin model, conditional on any valid construction history $\mathcal{H}$, with probability at least $1 - \alpha$ all catalogue contrasts satisfy*

$$D_i \in [l_i, u_i] := [\max\{A_i, \underline{m}_i - R_i(v_{i,L})\}, \min\{B_i, \bar{m}_i + R_i(v_{i,U})\}]. \quad (33)$$

The same coverage holds unconditionally. Subsequent selection among these covered contrasts, or evaluation at any deterministic implementation-price vector, needs no additional union bound. This conclusion applies only to policies and initial laws included in the catalogue; it does not create direct-cost observations for every unstudied tolerance or initial state.

The proof is in Supplement, Section 1.10.

In particular, selecting the first policy to meet a specified certificate is permitted when that selection uses construction records rather than the subsequent evaluation paths. Repeated construction clocks are not independent draws from a neural-optimizer population. The statistical randomness here belongs to the evaluation design. A fixed pseudorandom stream implements a reproducible experiment; it is not itself a proof of the independent-bin assumption.

10.2. *Replacement-fee identification*

Write $D = J^W - J^F$. If witness is installed, replacing it by FVI changes total cost by $k - D$, where $k \geq 0$ is the replacement fee. If FVI is installed, the change is $k + D$. Thus the two directional gains before fees are D and $-D$.

COROLLARY 15—The break-even fee frontier: *On an event where $D \in [l, u]$, the two directional gains belong to $[l, u]$ and $[-u, -l]$. The unknown absolute cost difference satisfies*

$$|D| \in [\text{dist}(0, [l, u]), \max\{|l|, |u|\}]. \quad (34)$$

For every $k \geq \max\{0, u, -l\}$ neither replacement strictly lowers expected cost. By contrast, $k \geq \max\{0, l, -u\}$ only removes a certified strictly profitable replacement. The two thresholds generally differ. Strict inequalities in the first criterion certify that both replacements strictly increase cost.

The proof is in Supplement, Section 1.11.

The fee frontier is more informative than a conclusion at one generous charge. The earlier 1/64 replacement decision is retained with its original interpretation and confidence account. The new analysis reports the two directional gain intervals and both thresholds instead of selecting a favorable fee after seeing the data. There is no welfare calibration hidden in the units: a decision maker must supply the economically relevant charge, or use the full frontier to see what remains unidentified.

10.3. *Observation costs on the same bit grid*

Let $\mathcal{B} = \{4, \dots, 16\}$ be the available bit precisions. For one fixed returned controller, write J_b for the expected cost of its actual b -bit, robustly repaired and quantized implementation. Use $b_0 = 16$ as a common reference and obtain simultaneous intervals for $D_b = J_b - J_{b_0}$. The identity interval at $b = b_0$ is exactly $[0, 0]$. Let $s_b = db \sum_{t < T} B_t$ be the discounted number of coordinate-bits purchased; the net cost is $J_b + \lambda s_b$.

THEOREM 16—Certified choice of actual net implementation cost: Suppose all $D_b \in [l_b, u_b]$ simultaneously. For any $\lambda \geq 0$ define

$$\begin{aligned} L(\lambda) &= \min_b \{l_b + \lambda s_b\}, & U(\lambda) &= \min_b \{u_b + \lambda s_b\}, \\ \widehat{b}(\lambda) &= \min \arg \min_b \{u_b + \lambda s_b\}. \end{aligned} \quad (35)$$

Then, simultaneously for all $\lambda \geq 0$,

$$0 \leq J_{\widehat{b}} + \lambda s_{\widehat{b}} - \min_{b \in \mathcal{B}} (J_b + \lambda s_b) \leq U(\lambda) - L(\lambda). \quad (36)$$

The lower and upper functions are finite piecewise-affine envelopes. All changes in their active lines occur at nonnegative intersections of catalogue lines, with exact comparisons deciding ties. The smallest selected precision $\widehat{b}(\lambda)$ is weakly decreasing in λ .

The proof is in Supplement, Section 1.12.

This result makes a statement about *actual* net cost within the declared bit catalogue. It differs from minimizing a uniform all-state loss bound. Both criteria are useful: the latter gives distribution-free protection relative to the full-information optimum, while (36) uses a specified initial law to compare feasible implementations of one returned policy. Neither criterion gives an optimum over arbitrary sensor technologies or all policies.

For a fair comparison the conventional actor receives its own acquisition allowance. Suppose its nearest-node policy has a certified selected-policy account at a reported state, obtained from the graph modulus L_t^{gr} , and its actual interpolated critic has modulus L_t^f . Transporting the same selected node action to the true state, then using cell-wide capacity repair, adds at most

$$(L_t^{\text{gr}} + L_t^f + 2D_t K_t^A) r(b) + D_t \Delta_t. \quad (37)$$

Indeed, transport the feasible node pair directly to the true state, bound its node distance by the reported-state distance plus $r(b)$, and bound $f_t(\widehat{x}) - f_t(x)$ by $L_t^f r(b)$. Robust repair adds $D_t(2K_t^A r(b) + \Delta_t)$. This proves (37) without assuming actor continuity. For a witness critic $L_t^f = L_t^{\text{gr}}$, it reduces to the preceding witness allowance. The empirical comparison therefore evaluates the all-state-account minimizer, the

interval upper-net-cost selector, and a descriptive midpoint selector on the *same* integer bit and price grid. The midpoint selector is not called the true optimum.

11. COMPLETE SERVICES AND ACTUAL POLICY COSTS

We evaluate the original nonlinear investment economy, not a finite-state replacement. The new design adds stronger prediction controls, prospective reuse, direct continuous-law policy inference and the existing two-control extension. The earlier tensor-reference and query studies remain separate evidence: in particular, the twenty strictly worse pure ReLU policies and the conventional winners of the earlier cold-start catalogue are not removed or reclassified.

11.1. *Frozen design and corrected accounting*

Two scalar tasks have $(d, T, \epsilon) = (2, 2, 1/128)$ and $(8, 3, 1/64)$. Each returned policy uses twenty-four-bit downward state acquisition, thirty-two-bit downward actions, the original law, and the same uniform original-optimum budget. The action-label catalogue has ninety-six states per seed. ReLU, full quadratic, linear, additive linear spline and nearest-action fits share those labels for each of eight fixed seeds. Adaptive parabolic search, bisection, the upper-chord minimizer, an analytic terminal-derivative proposal and a midpoint rule supply five unfitted controls. Every arm uses the same pre-query screen, protected refinement and terminal kernel.

Two process repetitions give 180 comparison services. Another 24 services train once and load the saved model for twenty-four new workloads. Two inference services compare four complete callable controllers on 8,192 common-path rows per task. Six vector services cover two state dimensions and three horizons. The resulting catalogue has 212 predeclared processes, with a 900-second deadline per process. A failed or timed-out process remains part of its catalogue and work account. Repeated clocks are not independent training seeds.

The present execution corrects an error found by an independent rational trajectory check. A point interval used to initialize accumulated cost shared its lower and upper arrays. Updating these arrays in place overwrote the running lower endpoint with an upper endpoint. The repair allocates distinct arrays; it changes

neither the Bellman kernels nor the policy-selection rule. We preserved the original execution and froze a complete corrected rerun with its own process receipts. The same random indices isolate the repair, but do not create another independent statistical sample. The active tables use only the corrected execution. Supplement, Section 2, gives the counterexample and the reproduction boundary.

11.2. Stronger predictors and observed certificate progress

The comparison charges complete process work from launch through shutdown and join. This includes imports, action labels, fitting, predictions, all Bellman queries, interval path evaluation, serialization and durable output. The terminal derivative is an analytic proposal in the original model, not an oracle for the preceding dates. Nearest action stores the finite training sample, not a policy on a tensor state grid. The upper-chord minimizer is supplied as an explicit conventional competitor rather than reserved for the learned arm.

All fitting seeds and warnings are reported. The table does not infer a population optimizer-success probability from a finite seed catalogue or a statistically identified runtime ranking from closely spaced clock measurements. The supplementary tables give every seed, every routed and declined proposal, and the separate changes in the upper witness and global lower bound. A routed query may improve neither endpoint; those observations remain in the progress account. The exact identity

$$(U_{\text{old}} - L_{\text{old}}) - (U_{\text{new}} - L_{\text{new}}) = (U_{\text{old}} - U_{\text{new}}) + (L_{\text{new}} - L_{\text{old}})$$

organizes the diagnostics. The margins in Theorem 11 describe the work relevant to certification; the fitted action-label mean squared error is reported separately and is not asserted to control this identity.

11.3. Measured reuse, including a declined model

The reuse catalogue fixes ReLU and quadratic seeds 6603 and 6607 before its workloads. Each service fits and serializes once, then loads and checks the model for every block. Block index 7 deliberately presents a stale acquisition-contract identifier. The model is declined and the same conventional verifier is used. The

TABLE I
TEN SELECTORS AT THE SAME ORIGINAL-OPTIMUM CONTRACT

d/T	Selector	Runs	Queries	Fit (s)	Full (s)	Warnings
2/2	Adaptive	2	1234	0.000	0.365	0
2/2	Bisection	2	1324	0.000	0.365	0
2/2	Upper chord	2	1274	0.000	0.365	0
2/2	ReLU	16	1209	0.368	0.766	6
2/2	Quadratic	16	1219	0.024	0.365	0
2/2	Linear	16	1224	0.023	0.365	0
2/2	Spline	16	1214	0.023	0.365	0
2/2	Nearest	16	1244	0.023	0.365	0
2/2	Derivative	2	1284	0.000	0.365	0
2/2	Midpoint	2	1289	0.000	0.365	0
8/3	Adaptive	2	14469	0.000	1.567	0
8/3	Bisection	2	17578	0.000	1.944	0
8/3	Upper chord	2	14913	0.000	1.643	0
8/3	ReLU	16	14569	0.919	2.620	32
8/3	Quadratic	16	14569	0.526	2.169	0
8/3	Linear	16	14638	0.523	2.144	0
8/3	Spline	16	14663	0.525	2.219	0
8/3	Nearest	16	14809	0.527	2.194	0
8/3	Derivative	2	15204	0.000	1.769	0
8/3	Midpoint	2	16383	0.000	1.793	0

Targets are $1/128$ for $d/T = 2/2$ and $1/64$ for $8/3$. Full time is the parent launch-through-join receipt.

Fitted selectors use eight declared seeds and two process repetitions; unfitted controls have two repetitions. Fit includes labels. Query and time entries are medians, not optimizer-success probabilities.

Warning counts retain every executed fit. All arms use the same outward pre-query screen.

file is not secretly repaired or silently accepted. Model loading, hashing, rejection, fallback, all queries and durable records belong to the measured service.

The supplementary crossing table compares each process with the adaptive process bearing the same repetition number. It reports the first observed cumulative crossing, any crossing that persists through the remaining observed blocks, and the final complete-process comparison. Prefix clocks end at a block's durable record and

TABLE II

PROSPECTIVE TWENTY-FOUR-BLOCK MODEL REUSE

d/T	Selector	Runs	Full (s)	Full range (s)	Wins	Declines
2/2	Adaptive	2	2.269	2.269–2.269	0/2	2
2/2	Bisection	2	2.420	2.370–2.469	0/2	2
2/2	ReLU	4	2.620	2.620–2.720	0/4	4
2/2	Quadratic	4	2.269	2.269–2.269	4/4	4
8/3	Adaptive	2	19.688	19.664–19.711	0/2	2
8/3	Bisection	2	22.766	22.665–22.866	0/2	2
8/3	ReLU	4	22.068	21.467–22.621	0/4	4
8/3	Quadratic	4	20.967	20.717–21.121	0/4	4

Full time includes imports, fitting, model serialization, all loads and identity checks, twenty-four new workloads, the stale-contract fallback, all durable records and process exit. Wins compare matched repetition numbers with adaptive search; trained methods retain both declared seeds. Each service declines block index 7. Prefix crossings are separately recorded and are not labeled full-process break-even times.

exclude the final summary and process exit; they are not mislabeled as complete costs of unexecuted stopping services. No crossing beyond the twenty-four-block window is extrapolated. This design replaces an assumed equal-query amortization ratio by actual repeated use, including unfavorable repetitions.

11.4. *Expected costs of the new callable policies*

The inference experiment conditions on the frozen adaptive, bisection, ReLU and quadratic controllers, with fitted seed 6603 fixed in advance. Initial states are uniform on the full state cube. Each uniform innovation retains its original continuous law. Forty-eight-bit bins represent intervals of true continuous states and shocks, not point-mass substitutes. If a propagated state interval straddles an acquisition boundary, the unresolved tail receives its full valid cost-support enclosure. Such a path is neither discarded nor evaluated at its midpoint for statistical inference.

Four absolute costs and six paired differences are evaluated at each task. The bounded sample-variance inequality allocates $1/4000$ to each of forty endpoint tails, giving family error $1/100$ across both tasks. The conservative logarithm bound is

TABLE III
EXPECTED COST OF THE ACTUAL CALLABLE CONTROLLERS

d/T	Controller	Expected-cost interval
2/2	Adaptive	[0.55849, 0.62153]
2/2	Bisection	[0.55849, 0.62153]
2/2	Quadratic	[0.55845, 0.62149]
2/2	ReLU	[0.55845, 0.62149]
8/3	Adaptive	[0.67503, 0.72574]
8/3	Bisection	[0.67503, 0.72573]
8/3	Quadratic	[0.67501, 0.72572]
8/3	ReLU	[0.67501, 0.72572]

There are 8,192 distinct IID-model common-path rows per task, with continuous 48-bit initial and innovation bins and full tail bounds on ambiguous acquisition. The simultaneous family error is 1/100 across both tasks, all absolute costs and paired differences. Endpoints are rounded outward. Repeated R66/R67 streams are counted only once.

nine, verified by an exact rational partial sum of the exponential series. There are 16,384 distinct common-path rows, not that number multiplied by the four methods or by the corrected rerun. A fixed pseudorandom stream ensures reproduction; independence is the explicitly stated bin-sampling model.

All twelve paired difference intervals contain zero. The table therefore supplies direct costs for the new complete callable policies without claiming that a change in query count identifies an economic gain. The lower and upper sample endpoints, negative paired gains and acquisition ambiguities are retained. The full paired intervals are in Supplement, Table II. The earlier cost comparisons concern different policies and are not substituted for these observations.

11.5. *Two controls, two innovations and horizon cost*

The vector services use the original state-dependent capacity, the nonseparable action interaction and two independent continuous innovations. Each evaluates four designed initial states whose coordinates are all zero, one, one quarter or three quarters. These are declared scaling paths, not a four-observation estimate of a

TABLE IV
COUPLED TWO-CONTROL, TWO-INNOVATION QUERY SCALING

d	T	Full (s)	Queries	Descendants	Batch	RSS (KiB)	Recovery
2	2	0.315	44	20	4	39348	0
2	3	7.380	2957	1224	16	39328	0
2	4	530.165	8106242	1894012	1024	43688	0
8	2	0.365	44	20	4	39336	0
8	3	11.491	2957	1224	16	39528	0
8	4	841.366	8106112	1893976	1024	44836	0

Each service evaluates four designed initial states (all coordinates 0, 1, 1/4 or 3/4), with continuous-bin innovations, at target 1/4. No state lattice or trained two-output selector is loaded. Batch is the largest recorded query batch, not the total number of live descendant states. RSS is process peak memory. Horizon-four cost is retained rather than extrapolated from the shorter horizons.

population policy cost. A triangular action cover replaces the scalar action interval; no state lattice or stored reference policy is consulted.

The horizon-four rows make the descendant cost visible. Query batches and peak resident memory are distinct records: the algorithm can allocate descendants before batching their evaluation. The uniform policy guarantee follows from the theorem and its precision conditions, not from extrapolation from four paths. This experiment executes the multivariate-control verifier; it is not a trained two-output neural comparison or a demonstration that long-horizon work is negligible.

11.6. The purchase decision

For a candidate complete service j and benchmark b , let $[l_j, u_j]$ be the simultaneous interval for $J^b - J^j$, let ΔW_j be its measured work difference, and let f_j be the incremental adoption fee. If $v \geq 0$ values one normalized unit of expected cost and $p \geq 0$ prices one unit of measured work, the net-benefit interval is

$$[vl_j - p\Delta W_j - f_j, vu_j - p\Delta W_j - f_j]. \quad (38)$$

A positive lower endpoint licenses adoption; a negative upper endpoint rules it out; otherwise the decision remains unresolved on the stated information. Applying this

criterion to complete reuse receipts is different from treating a prefix crossing as a completed economic service. For a purchaser requiring only the stated accuracy contract, the lowest measured complete work among eligible services is a separate finite-catalogue criterion. Both are expressed in the original normalized units. No unobserved installation price, dollar calibration or empirical welfare magnitude is introduced.

12. CONCLUSION

Neural Bellman Operators connect continuation construction to feasible decisions under the original economic law. The witness construction supplies an explicit own-future loss account. Primitive curvature also supports a recursive query implementation without a state lattice. A computable upper chord can license a predicted action before another Bellman evaluation, while the lower certificate retains authority over accuracy.

The operational loss measures verification work rather than distance to an unknown optimum. Its fixed-feature ReLU readout has a certified convex optimization account. The two-control query extends the same policy guarantee to the previously specified coupled-capacity economy. These are constructive statements, distinct from a claim that a particular fitted network wins a computational comparison.

The full study makes that comparison observable. It charges training, loading, invalid reuse, queries, path enclosures and durable records; evaluates actual complete policies under the continuous law; and records the growth of a two-innovation continuation tree. The current gain intervals do not identify a representation-specific economic advantage. Keeping that finding alongside the new positive certificates preserves the paper’s original objective: a useful numerical decision must be feasible, accurate and worth its complete cost.

REFERENCES

- AGRAWAL, AKSHAY, SHANE BARRATT, STEPHEN BOYD, AND BARTOLOMEO STELLATO (2020): “Learning Convex Optimization Control Policies,” in *Proceedings of the 2nd Conference on Learning for Dynamics and Control*, 120, 361–373. [3, 19]
- FELZENSZWALB, PEDRO F. AND DANIEL P. HUTTENLOCHER (2012): “Distance Transforms of Sampled Functions,” *Theory of Computing*, 8 (19), 415–428. [3, 8]

- HOEFFDING, WASSILY (1963): “Probability Inequalities for Sums of Bounded Random Variables,” *Journal of the American Statistical Association*, 58 (301), 13–30. [27]
- MC SHANE, EDWARD J. (1934): “Extension of Range of Functions,” *Bulletin of the American Mathematical Society*, 40 (12), 837–842. [3, 7]
- MUNOS, R. (2003): “Error Bounds for Approximate Policy Iteration,” in *Proceedings of the Twentieth International Conference on Machine Learning*, 560–567. [3]
- PIROTTA, MATTEO, MARCELLO RESTELLI, ALESSIO PECORINO, AND DANIELE CALANDRIELLO (2013): “Safe Policy Iteration,” in *Proceedings of the 30th International Conference on Machine Learning*, vol. 28 of *Proceedings of Machine Learning Research*, 307–315. [3]
- SAKAUE, SHINSAKU, AND TAIHEI OKI (2023): “Rethinking Warm-Starts with Predictions: Learning Predictions Close to Sets of Optimal Solutions for Faster L -/ $L^{\natural}$ -Convex Function Minimization,” in *Proceedings of the 40th International Conference on Machine Learning*, 202, 29760–29776. [3, 19, 29]
- SAMBHARYA, RAJIV, GEORGINA HALL, BRANDON AMOS, AND BARTOLOMEO STELLATO (2023): “End-to-End Learning to Warm-Start for Real-Time Quadratic Optimization,” in *Proceedings of the 5th Annual Learning for Dynamics and Control Conference*, 211, 220–234. [3, 29]
- SCHERRER, BRUNO, MOHAMMAD GHAVAMZADEH, VICTOR GABILLON, BORIS LESNER, AND MATTHIEU GEIST (2015): “Approximate Modified Policy Iteration and its Application to the Game of Tetris,” *Journal of Machine Learning Research*, 16 (49), 1629–1676. [3]
- YANG, INSOON (2020): “A Convex Optimization Approach to Dynamic Programming in Continuous State and Action Spaces,” *Journal of Optimization Theory and Applications*, 187, 133–157. [3]